MIT Art Design and Technology University

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Title: Inventory Management System

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Introduction

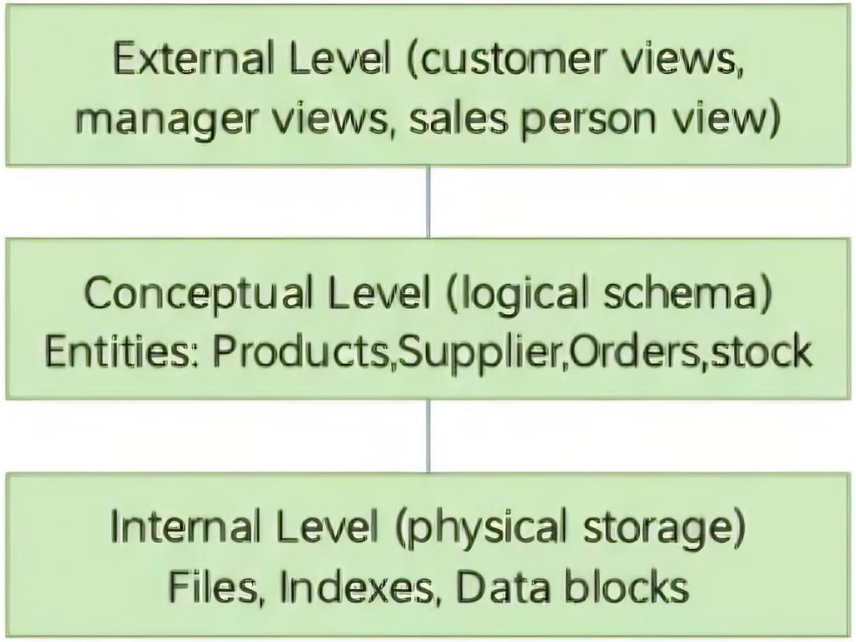
- **Definition:** An Inventory Management System (IMS) is a software/database solution used to track, manage, and control stock levels.
- **Purpose:** Ensures efficient handling of products, suppliers, customers, and orders.

- Provides real-time updates for better decisions.
- DBMS ensures accuracy, security, and easy access to data.
- Improves efficiency and reliability in inventory control.

- **Outcome:** IMS with DBMS improves efficiency, accuracy, and scalability in managing inventory.

Database Architecture

- External Level (user views like manager, supplier, customer).
- Conceptual level (tables & relations)
- Internal Level (physical storage)



Database Architecture

- **1. External Schema (User Views)**
- This is the **top level**, closest to users.
- It shows **different views for different users** depending on their role.
- In IMS:
  - **Inventory Manager View** → Can see product stock levels, warehouse status, low-stock alerts.
  - **Procurement Officer View** → Can see supplier details, purchase orders, pending deliveries.
  - **Sales Manager / Customer View** → Can see customer orders, invoices, and sales reports.
  - **Admin View** → Full access to all data (Products, Suppliers, Customers, Orders, Stock).

- The Inventory Manager's screen only shows product stock details, not customer or supplier details.

Database Architecture

- **2. Conceptual Schema (Logical Schema)**
- Shows the **entire database design logically**.
- Example tables/entities:
  - **Products** → ProductID, Name, Category, Price, StockLevel
  - **Suppliers** → SupplierID, Name, ContactInfo
  - **Orders** → OrderID, SupplierID, Date, TotalAmount
  - **Inventory** → ProductID, Warehouse, Quantity

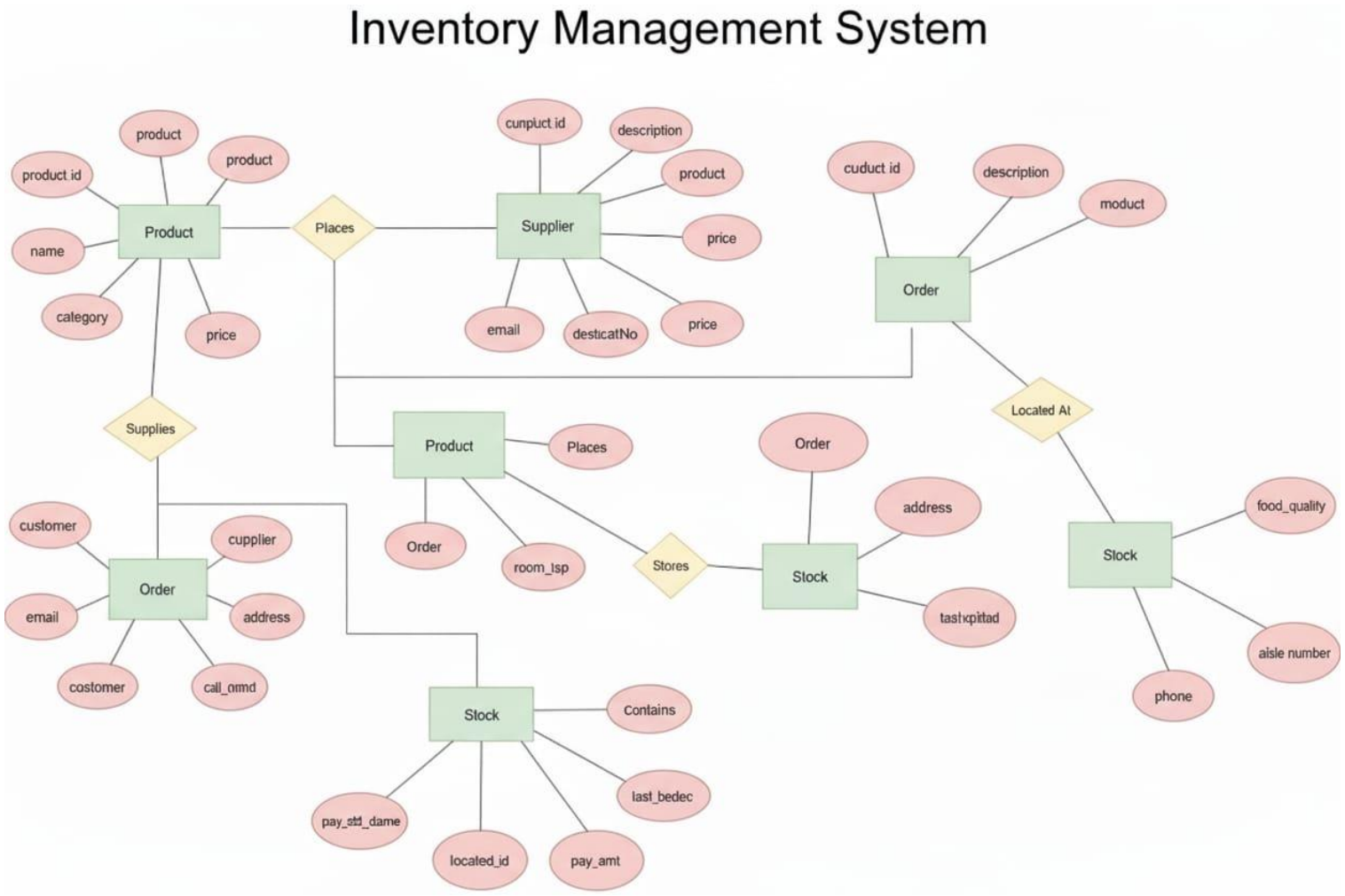
- At this level we know "A Supplier supplies Products" and "Orders include Products," but we don't see *how* they are stored physically.

Database Architecture

- **3. Internal Schema (Physical Schema)**
- Describes **how data is actually stored** in the database system.
- Example in IMS:
  - Product details stored in data files on disk.
  - Index built on **ProductID** for fast product lookup.
  - Orders stored in separate partitions by date.
  - Backup storage ensures recovery of inventory records.

- When you search for a product in IMS, the DBMS fetches it quickly using indexes — but users don’t see this process.

ER Diagram


Inventory Management System
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Tables

Product	Supplier	Customer	Orders	Stock
Product_ID (PK)	Supplier_ID (PK)	Customer_ID (PK)	Order_ID (PK)	Stock_ID (PK)
Name	Name	Name	Customer_ID (FK) → Customer.Customer_ID	Product_ID (FK) → Product.ID
Category	Email	Email	Supplier_ID (FK) → Supplier.Supplier_ID	Warehouse_Location
Price	Contact_No	Phone	Product_ID (FK) → Product.Product_ID	Quantity
Stock_Level	Address	Address	Order_Date	Aisle_Number
-	-	-	Quantity	Food_Quality
-	-	-	Total_Amount	Phone

Query 1 – Restocking Products

- Operation: Find products that need immediate restock.

- Relational Algebra:

$\sigma(\text{StockLevel} < 10)\ (\text{Products})$

- SQL:

```sql
SELECT ProductID, Name, StockLevel
FROM Products
WHERE StockLevel < 10;
```

Output

ProductID	Name	StockLevel
P101	Laptop	5
P202	Keyboard	3

Query 2 – Supplier Performance

- Operation: Rank suppliers by supply value.

- Relational Algebra:

π(SupplierID, ProductID, SUM(Quantity)) (Orders $\bowtie$ Products)

- SQL:

```sql
SELECT s.SupplierID, s.Name, o.OrderID, o.TotalAmount
FROM Suppliers s
JOIN Orders o ON s.SupplierID = o.SupplierID;
```

- Ouput

SupplierID	Name	OrderID	TotalAmount
S01	Tech Distributors	O501	5000
S02	Prime Supplies	O502	3000
S01	Tech Distributors	O503	7000

Query 3 – Common Items Between Warehouse

- Operation: Find overlapping items in warehouses A and B.

- Relational Algebra:

(Inventory[Warehouse=A] ∩ Inventory[Warehouse=B])

- SQL

```sql
SELECT ProductID, Quantity
FROM Inventory
WHERE Warehouse = 'A';
```

- Output

ProductID	Quantity
P101	50
P202	30
P303	20

Query 4 – Products with Orders (Inner Join)

- Operation: List ordered products.

- Relational Algebra:

Orders ⋈ Products

- SQL:

```sql
SELECT o.OrderID, p.Name, o.Date
FROM Orders o
JOIN Products p ON o.ProductID = p.ProductID;
```

OUPUT

OrderID	Name	Date
O501	Laptop	2025-08-10
O502	Keyboard	2025-08-12
O503	Mouse	2025-08-13

Query 5 – All Products by Category

- Operation : Inventory value by category

- Relational Algebra:
γ(Category, SUM(Price*StockLevel))(Products)

- SQL Query:

```sql
SELECT ProductID, Name, Category, Price, StockLevel
FROM Products;
```

- OUPUT

ProductID	Name	Category	Price	StockLevel
P101	Laptop	Electronics	50000	5
P202	Keyboard	Electronics	2000	3
P303	Pen	Stationery	10	100

Aggregation & Grouping

- Total stock of each product

```sql
SELECT SUM(QuantityAvailable) AS TotalStock
FROM Inventory;
```

- Total orders per customer

```sql
SELECT COUNT(OrderID) AS TotalOrders
FROM Orders;
```

- Average order quantity of products

```sql
SELECT AVG(Quantity) AS AvgOrderQty
FROM Orders;
```

Joins

- INNER JOIN: Get all orders with product details

```sql
SELECT Orders.OrderID, Customers.Name AS CustomerName,
       Products.Name AS ProductName, Orders.Quantity
FROM Orders
INNER JOIN Customers ON Orders.CustomerID = Customers.CustomerID
INNER JOIN Products ON Orders.ProductID = Products.ProductID;
```

- LEFT JOIN: Show all products even if not ordered

```sql
SELECT Products.Name AS ProductName, Orders.OrderID
FROM Products
LEFT JOIN Orders ON Products.ProductID = Orders.ProductID;
```

- RIGHT JOIN: Show all orders even if product details missing

```sql
SELECT Products.Name AS ProductName, Orders.OrderID
FROM Products
RIGHT JOIN Orders ON Products.ProductID = Orders.ProductID;
```

Set Operations

- **UNION**

Combines results from two queries into **one list (no duplicates)**.

```sql
SELECT Name FROM Products
UNION
SELECT Name FROM Suppliers;
```

- **INTERSECT**

Returns only the **common values** between two queries.

```sql
SELECT Name FROM Customers
INTERSECT
SELECT Name FROM Suppliers;
```

- **EXCEPT (or MINUS)**

Returns values from the **first query that are not in the second**.

```sql
SELECT ProductID FROM Products
EXCEPT
SELECT ProductID FROM Orders;
```